

Isabel Allum

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EDUCATION

Columbia University Dec 2026 (Expected)
MS in Computer Science, Machine Learning Track (GPA: 4.00) New York, NY
Courses: Machine Learning - Reinforcement Learning - Artificial Intelligence - Natural Language Processing

University of Chicago 2019-2023
BA in Economics, Specialization in Data Science (GPA: 3.89) Chicago, IL
Honors: Cum Laude, Dean's List (2020, 2022)
Courses: Econometrics and Machine Learning - Big Data - Data Science Clinic - Data Construction and Interpretation

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, scikit-learn), R (tidyverse, ggplot2), SQL, Stata
Data Tools/Platforms: PySpark, Databricks, Git/GitHub, Google Cloud Platform
Methods: Regression, Causal Inference (DiD, RD, IV), Clustering (k-means), Tree-Based models, Time Series Analysis

EXPERIENCE

Machine Learning Researcher Feb 2026 - Present
Columbia University (Complex Resilient Intelligent Systems Laboratory) New York, NY
• Developing a multi-agent reinforcement learning model of supply chain interactions to simulate equilibrium resilience

Economic Data Associate Jul 2023 - Jun 2025
Charles River Associates Chicago, IL
• Conducted statistical modeling and end-to-end data analysis on terabyte-scale datasets using PySpark (Databricks), R, and Stata, supporting 10+ multi-million dollar antitrust and competition litigation matters
• Built reproducible data validation and feature engineering workflows to generate analysis-ready datasets from raw client transaction data and public economic data
• Applied econometric methods (regression modeling, panel analysis, robustness checks, hypothesis testing) to quantify competitive effects and to evaluate “but-for” counterfactual scenarios
• Communicated complex technical findings for a diverse audience through clear visual exhibits and written reports; ensured analytical defensibility under legal scrutiny
• Reviewed junior analysts’ code and exhibits to ensure accuracy, consistency, and methodological soundness

Research Assistant Jun 2021 - Sep 2021
University of Chicago (Booth School of Business & Harris School of Public Policy) Chicago, IL
• Engineered a data simulation pipeline in R to process individual-level datasets, algorithmically applying 25+ complex unemployment insurance frameworks to quantify coverage gaps and user impact under various proposals
• Findings published by the American Economic Association and presented to Congress to guide national legislation

PROJECTS

Dinner Party Planner Web Application Sep 2025 - Dec 2025
• Designed and implemented a normalized PostgreSQL database schema to efficiently store and query user, event, recipe, and rating data with complex relationships across multiple tables
• Built a Flask-based web app on Google Cloud Compute Engine to enable real-time data capture of user interactions, utilizing SQL queries to handle operations like dynamic rating updates, recipe search, and event management

Perpetual (Data Science Consulting) Jan 2023 - Mar 2023
• Co-developed a spatial optimization model using weighted k-means clustering in Python to identify efficient distribution and collection hubs for reusable foodware in 2 cities; Presented recommendations to nonprofit leadership
• Integrated external demand signals using data from Google and Yelp APIs, performed data cleaning and feature engineering, and visualized geographic demand patterns